

Introduction; Solar energy-thermal and photovoltaic

What is energy?

What is the need for energy resources?

The term energy is derived from the Greek word en-ergon meaning that in-work

To perform any work in this world we require energy.

Let's take an example

To cook our food, we need energy.

To travel from one place to another place we need energy.

To keep vegetables and fruits afresh inside the refrigerator we need energy

To heat our house, we need energy

To cool down our rooms we need energy

To work with our mobile phones and laptop we need energy.

We are using some source of energy for a variety of applications.

If you take the history humans started using firewood and plant materials for cooking their foods.

Humans used animals for transportation such as a horse for riding

During the last century breakthroughs have happened. That is Benjamin Franklin discovered the electricity.

Followed by him Michael Faraday invented the electric dynamo the machine which converts mechanical energy into electricity in the year 1831

Thomas Alva Edison invented the light bulb which will glow when electricity passes

Nikola Tesla discovered the AC current motor which would convert electrical energy into mechanical energy.

We can't even imagine life without fuel or electricity, right?

The energy consumption of a country is usually considered as a sign of its development.

Let's discuss about

Types of Energy Resources

Energy resources are broadly classified into renewable or non-renewable energy resources.

Non-renewable energy resources are exhaustible and cannot be renewed. It can be available only for a limited period of time. Once they are consumed soon, they will run out.

On the other hand, Renewable energy resources are inexhaustible and they can be renewed. They are replaced as fast as they are used.

Can you say some renewable and non-renewable energy resources?

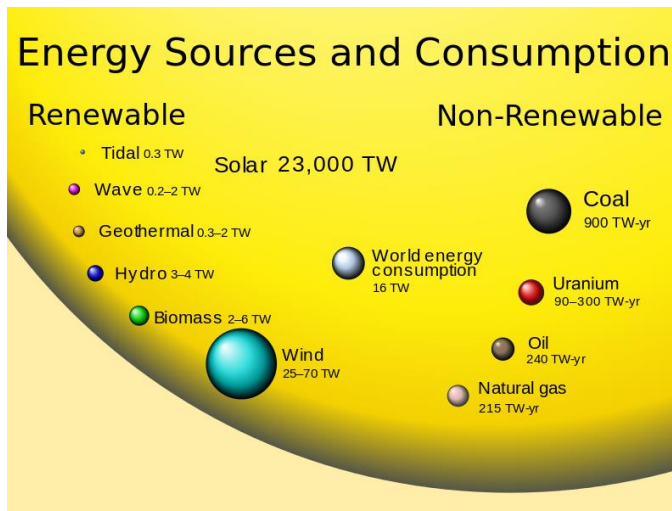


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Non-renewable Energy Resources are Fossil fuels such as oil petroleum diesel coal and natural gas.

The drawback with fossil fuels is if fossil fuels are burned to generate energy, they may release pollutants or carbon dioxide or other greenhouse gases into the atmosphere which will lead to global temperatures rising. It is called global warming.

Next Renewable Energy Resources

Examples of renewable energy resources are solar energy geothermal energy wind energy biomass hydroelectric power and tidal energy.

These energy resources are inexhaustible and can be renewed or replaced faster than we can use them.

Important Things to Consider about Energy Resources

Whether renewable or non-renewable energy resources two important things must be considered.

The first one is successfully making a useful form of energy from the energy resource. For example, how to harvest electricity from the sun.

Another one is Net Energy Net energy is the amount of high-quality energy produced from an energy resource minus the amount of energy required to develop it.

Let's discuss this by taking one example

if we get much less energy as output by burning fuel than the input energy to produce it. Then That particular fuel is probably not a practical energy resource. For example, nuclear power.

In order to produce nuclear energy, we need to spend a lot of energy on mining and extraction of uranium isotopes and enrichment of uranium fuels and fuel rod fabrications. Further after electricity is generated from uranium, we need to spend a lot of money on the storage of used fuel rods safely. Thus, net energy will be low

On the other hand, for another fuel net energy is high but if it creates large amounts of pollution that particular fuel also may not be the best choice for an energy resource. For example, a coal-based thermal power plant produces a lot of energy. Net energy will be higher than the nuclear energy. However, if you consider the amount of CO₂ release or

greenhouse gas emission is very high in coal-based power plants. Thus, even though net energy is high in coal-based thermal energy this is not an eco-friendly energy resource.

In this module we are going to learn more about renewable energy resources

First, we can start with SOLAR ENERGY.

Sun is an abundant source of energy, and it is inexhaustible.

Solar energy actually supports all life on earth

Thus, directly or indirectly the sun is the source of all the energy available on earth.

Solar energy applications can be classified into two categories one is direct solar energy and another one is indirect solar energy.



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Plants need sunlight to grow.

Animals and humans were dependent on plants for their food and for oxygen.

Once plants and animals die, they will reach underground. After millions of years later these dead plants and animals turn into fossil fuels such as coal and oil. Hence coal and oil are regarded as indirect solar energy.

Solar energy causes pressure differences in the atmosphere and this causes the movement of air that is wind. So, wind energy is actually a byproduct of solar energy.

Let's take the water cycle Water evaporates because of heat condenses to form clouds and precipitates back to earth in the form of rain and snow. This water can be stored in dams to produce hydroelectricity. Hence hydroelectricity is an indirect form of solar energy.

Now we came to an important discussion that is Direct solar energy usage

Direct solar energy can be obtained using two methods. The first one is Thermal example is Solar radiation can be absorbed in solar collectors to provide hot water.

The second one is Photovoltaic. For example, solar energy can be converted directly into electricity using photovoltaic panels which are normally mounted on roofs.

Let me explain Direct solar energy applications

Solar energy is abundant found everywhere and has no political barrier everlasting and available for free of cost

Solar energy is one of the cleanest and most easily accessible sources of energy.

Direct solar energy can be harvested in various ways; there are three ways to harness solar energy

First one is passive solar energy system

Second one is active solar energy system

Third one is photovoltaic.

First, we can discuss about how we can utilize Solar thermal heat for various applications

It is not a new concept

Solar thermal heat was used to evaporate seawater to produce salt

Solar thermal heat was used to dry our food

Solar thermal heat was used to dry our clothes

These are a few examples of passive solar energy applications.

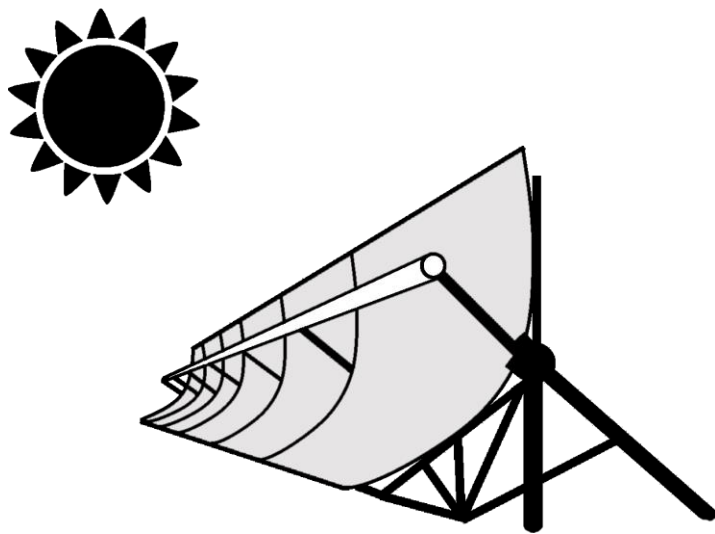


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The best example of a passive solar energy system is a solar cooker which is a device that uses sunlight to cook food.

Solar cookers work without any large complex systems of lenses or mirrors

We all know that when sunshine falls on a dark surface or black colored surface it absorbs solar energy, and it heats up.

The solar cooker uses the same principle.

Inside a solar cooker a glass-covered chamber is painted black, and the entire unit will be insulated. When we keep the solar cooker in sunlight it absorbs solar energy and heats up the surrounding air. This warm air circulates throughout the box and cannot escape. Because of this sometimes the inside temperature would reach more than 100 °C which is sufficient to cook our food.



Photo courtesy flickr.com

However Solar cooker takes longer times to cook food. This is the best example of a passive solar energy system

In most parts of India where solar radiation is relatively abundant; thus, we can use solar box cooker to prepare food more sustainable way

In India we have the largest solar steam cooking system. More importantly it is still working from the 90s in Brahmakumaris Ashram at Mount Abu in Rajasthan.

They have 84 shining parabolic concentrators on the roofs; each one looking like a huge dish made of reflecting concave mirrors. The sunlight from the concentrators heats up the receiver and converts water into steam. This system can cook for more than 38000 people.

The next example of Passive use of solar energy is daylighting

Today many buildings are designed to take advantage of natural solar energy for daylighting. Daylighting is simply the use of natural sunlight to light up a building's interior.

The south side of a building mostly receives the highest sunlight Therefore buildings are designed south facing for passive solar daylighting. Usually, the building is designed with large south-facing windows. This kind of design will allow the entry of maximum sunlight into the building's interior.

Passive solar systems are maintenance-free. There are no operating costs. We can substantially reduce the electric bills.

There is no need for devices such as external pumps fans or electricity.

The only major problem is passive solar heating or passive solar lighting systems depend on the climate.

In a cloudy and dark climate, it will not operate.



Photo courtesy flickr.com

The second method of solar energy application is Active solar energy utilization

Active solar energy utilization means capturing and storing of solar energy for future use.

In active solar heating systems solar energy will be used to heat up fluid or air and then the fluid is moved with the help of external pumps to the storage system and then the captured heat will be transferred directly for later use.

Usually, active solar heating and solar cooling systems require solar collectors which are usually mounted on roofs.

Such systems also require pumps and motors to move the heated fluids to the storage system in order to deliver the captured heat.

Solar water heaters are the best example of active solar energy utilization. Typical solar water heaters consist of two parts one is solar collector and the second one is storage tank.

First, we can discuss about Solar Collectors.

A typical water heater is composed of solar flat plate collectors.

Solar radiation is absorbed by the collector and converts the incident solar radiation into thermal energy by absorbing heat. The heat gathered is transferred to the storage unit.

The solar collectors are usually placed on the roof of the building facing south and at an inclination of 30-60 degrees with respect to the horizontal plane.

Solar collectors are classified into two categories. The first one is Non concentrating collectors and the second one is Concentrating collectors.

Let me explain what Non concentrating collectors is

Non concentrating collectors means the area that intercepts the sunlight is the same as the area absorbing the solar energy

Flat-plate collectors are the most common example of non-concentrating collectors which are used widely for water heating.

These collectors are simply metal boxes that have a transparent glass cover on top of a dark or black-colored absorber plate

The rest of the faces of the box are insulated to prevent heat losses. These boxes consist of copper pipes running in parallel. The fluid typically water flows through these copper pipes.

Solar radiation passes through the transparent glass material and absorbs the radiation from the sun. The circulating water inside copper pipes heats up and transfers the heat to water in a storage vessel.

This can be used for domestic purposes as a water heater or heating the water in the swimming pools in warm climates.

Next, I will explain about Concentrating Collectors

In Concentrating collectors, the area intercepting solar radiation is greater than the absorber area.

Solar Furnace and Solar thermal power plants use concentrating solar collector systems.

A solar furnace is an optical system that uses concentrated solar power to produce high temperatures. The solar furnace technique is based on reflecting solar radiation from Parabolic mirrors or heliostats and concentrating it onto a focal point.

The largest solar furnace was installed at Mont-Louis in France. it has been operational since 1970.

Nearly 20000 mirrors were used to concentrate sunlight to create more than 3500 °C

Temperature at the focal point of this solar furnace



Photo courtesy flickr.com

Next example for active solar energy utilization is Solar energy to produce electricity

Solar energy is used to generate electricity.

Solar collectors in sunny deserts can produce high-temperature heat which drives a heat engine nothing but a steam turbine which is connected to an electrical power generator for producing electricity.

The advanced computer-connected solar collectors usually move by tracking the sun to maintain a high degree of concentration on a central heat collection unit and transform solar energy received from the sun into high-temperature heat energy which can be used to convert heat energy into electricity.

Next, we can see Solar energy for cooling applications

The principle behind the solar cooling or solar refrigeration is like conventional refrigeration. But the difference is Solar thermal energy is used instead of electrical power to operate a heat engine.

The heat engine compresses a special vapor into a liquid refrigerant.

The re-evaporation of this liquid refrigerant absorbs the heat out of the surroundings and in turn cools its surroundings.

Final topic I want to discuss is Solar cells or photovoltaic technology

Modern solar power systems use photovoltaic cells to collect solar energy.

“Photo” means “produced by light” and “voltaic” is “electricity produced by a chemical reaction.”

Simply the process of converting photons into electricity.

A single PV device is commonly called as a solar cell.

The photovoltaic cell contains a semiconductor most commonly silicon with a small number of impurities such as boron or phosphorus. Recently PV cells are made up of cadmium telluride or copper indium gallium diselenide. Arsenic and antimony have also been used in solar cells. New photovoltaic technologies are coming up such as solar cells made up of organic materials or quantum dots or made up of hybrid organic-inorganic materials perovskites materials such as calcium titanate.

A typical solar cell contains a very thin semiconductor and is often less than the thickness of human hairs.

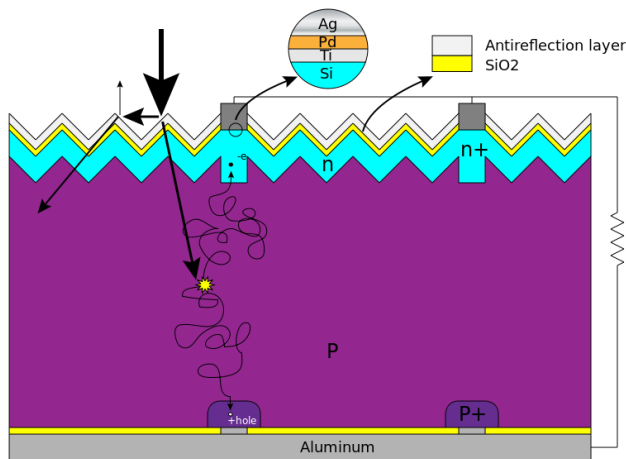


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Each cell is connected by a circuit and designed into modules or panels. Several panels can be connected to form arrays. One or more arrays are then connected to the electrical grid.

When sunlight falls on the silicon layer it causes electrons to eject. And these ejected electrons move quickly into the circuit and generate electricity.

Commercial or domestic PV panels produce an average current from 10 watts to 300 watts in a direct current.

PV panels require an inverter to change the DC electricity into AC current in order to be compatible with electrical devices and the electric grid.

PV panels can **also** be used to create large-scale power plants. Bhadla Solar Park is the world's largest solar park which is located in Rajasthan India.

It is spread over a total area of 14000 acres and generates 2250 Mega Watts of electricity

PV cells can be used to power space satellites

Solar cells can be used. to provide electricity to remote villages street lighting applications. in the desalination of salt water and water pumping and so on.

PV cells are used Powering remote telecommunication devices and railway signals.

Also powering of smaller items such as calculators and watches.